

Toward Optimized AR-based Human-Robot Interaction Ergonomics: Modeling and Predicting Interaction Comfort

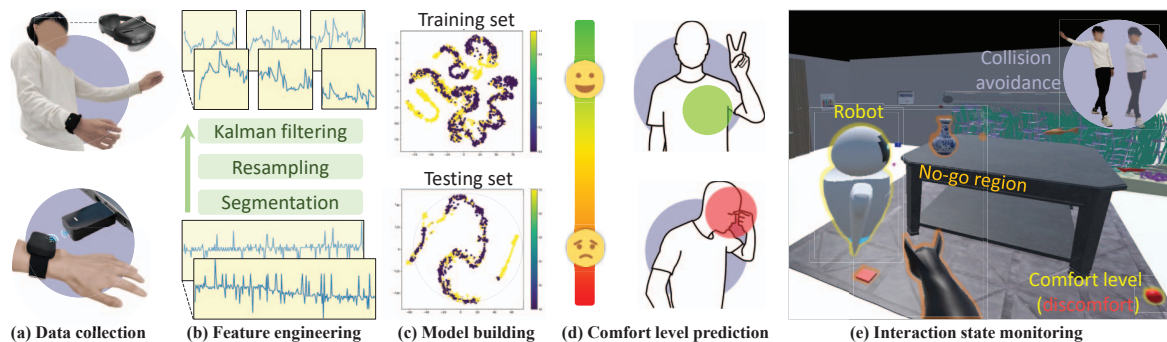
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Figure 1: The steps of developing a comfort monitoring model.

ABSTRACT

Augmented Reality in Human-Robot Interaction (AR-HRI) boosts user experience. The key challenge is refining interaction methods to minimize discomfort and enhance quality. This AR-HRI study uses Galvanic Skin Responses (GSR) to predict and improve user comfort. User studies tested interaction strategies in an AR environment. A machine learning model, developed from GSR data, predicted comfort levels and informed strategy changes. Comfort metrics were visualized every second using Hololens 2, creating an AR-HRI comfort system. The method improved user comfort, provided a new AR-HRI metric, and highlighted future research opportunities.

Index Terms: Human-centered computing—Interaction design

1 INTRODUCTION

Augmented reality enhances human-robot interaction (AR-HRI) by offering benefits such as easy shape modification, enhanced feedback, and improved safety to the robot [3]. However, selecting the right interaction method remains a challenge. Current methods offer multiple methods at task start [5], instruct fixed methods during tasks [1], or auto-switch methods based on interaction quality [2]. These methods have limitations, such as not considering direct physiological signals or predicting comfort in advance. The impact of these methods on AR-HRI comfort needs further investigation. In this paper, we introduce a method to predict interaction comfort in AR-HRI using GSR signals, which are indicative of discomfort. Additionally, we develop a comfort monitor, updated every second, visualized with Hololens 2. We collect GSR and subjective evaluation data from user studies and build machine learning models for comfort prediction. When the system detects discomfort, it recommends adjustments to the interactions.

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2 METHOD

2.1 Interaction Comfort Level in HRI

This paper proposes a method to predict interaction comfort in AR-HRI using GSR signals and the NASA-TLX Load Index. We conducted a study with 12 participants, who wore a GSR sensor wristband and a Hololens Fig. 1(a). They performed tasks involving moving obstacles and adjusting robot movement routes with different interaction methods (gesture, gaze, or voice+gaze). We collected GSR signals and comfort ratings from the participants, and processed the GSR signals using segmentation, resampling, and Kalman filtering. We used wavelet features of GSR signals to compare the comfort levels of different groups, and found significant differences by using one-way analysis of variance (ANOVA, non-parametric method, Kruskal-Wallis). We're developing a GSR-based model for AR-HRI to predict and alert discomfort, promoting more comfortable interactions. Our research questions are:

(1) **RQ1:** How do GSR signals affect the interaction comfort level in AR-HRI under different interaction methods?

(2) **RQ2:** How does the interaction comfort level monitoring model influence the comfort of participants' interaction experiences?

2.2 Modeling and Predicting Interaction Comfort Level

We develop models to link GSR signals to comfort levels and use supervised learning to detect cognitive states from physiological features. We process the raw GSR data with feature selection and normalization, and split it into training and validation sets (8:2). The dataset, with 10,000 equally distributed "discomfort" (represented by 0) and "comfort" (represented by 1) labels, uses physiological time series signals as inputs. Fig. 1(b-d) shows the prediction model development steps. We use precision and F1 score to evaluate model performance. Table 1 compares various algorithms. The optimal KNN model, chosen for its accuracy and bias-variance balance, is used for training. Finally, discomfort predictions are sent to Hololens via UDP (User Datagram Protocol). If discomfort is detected, a red light (Fig. 1(e)) prompts the user to adjust their interaction method.

3 EVALUATION

3.1 Indoor HRI Collision Avoidance Experiment

Participants and setup. 31 participants (21-28 years old, 6 females) in two groups. Group A used a comfort monitor, Group B didn't. All wore a Hololens 2 and a GSR wristband, and performed an path adjusting task. The **AR-based environments.** Combined real indoor scenes with virtual assets. The virtual robot showed possible actions and helped intention alignment [3]. **Tasks for adjusting the robot's path.** The robot cleaned the house, avoiding fragile items and no-do regions. Humans modified the robot's path. **Interaction methods.** Included gesture, gaze, and voice+gaze. The human could drag the robot. **Task duration.** 5 sessions (5 minutes each) with breaks. Each session had a unique scene and 4-6 targets in different rooms. Interactions per target lasted 5-8 seconds. **AR-HRI comfort monitoring system.** Group A utilized a comfort monitor equipped with AR indicators. The system alerted the user about discomfort and promoting more comfortable interactions Fig. 1(e). **Interaction comfort level evaluation.** Groups A and B's comfort was evaluated using subjective and objective methods. Objective data included interaction duration, "select" and "drop" actions, and robot collision failures. Subjectively, participants used the NASA-TLX (2 items: Physical Demand and Frustration Level).

3.2 Result

We compared data from Group A, who used the comfort monitor, and Group B, who did not. The data included interaction duration, number of actions, robot collision failures, and comfort ratings. Having passed the normality and homogeneity of variance tests, we report these results and conduct an independent t-test to identify significant group differences. **Duration of Interaction in a Single Session.** Group A interacted for an average of 281.56s (SD = 16.190), Group B for 284.67s (SD = 12.33). No significant difference was found. **Aggregate Count of "Select" and "Release" Interactions.** Significant difference found by t-test, $t(29) = -3.72, p < .001$. Group A ($M = 16.00, SD = 4.69$) interacted less than Group B ($M = 23.13, SD = 5.92$), suggesting better alters. Some users (P8, P21 and P25) preferred voice over gestures when tired or stressed. **Incidents of Robot Collision Failures.** Group A ($M = 10.38, SD = 3.22$) had fewer failures than Group B ($M = 15.00, SD = 1.81$), suggesting effective alerts. Significant disparity found, $t(29) = -4.87, p < .001$. **Perceived Comfort During Interaction.** Group A, on average, rated their comfort at 4.25 (SD = 0.57), whereas Group B's average rating was 3.20 (SD = 0.41). The t-test highlighted a significant difference between the groups, $t(29) = 5.78, p < .001$.

4 DISCUSSION

The comfort monitor can improve user interactions, robot navigation efficiency, and user satisfaction. Further exploration could examine the specific features of the system that are most impactful.

4.1 Applications and Benefits

The combination of discomfort alerts and AR-HRI tasks, such as path adjusting, offers novel applications and advantages in human-robot interaction. **Predictive Actions and Safety Confirmation.** Predicting human actions, a task both challenging and safety-critical,

is enhanced by comfort prediction, enabling predictive and active behaviors. This is particularly vital in dynamic scenarios such as autonomous driving, where the system actively works to prevent collisions. **Mixed Human-Robot Collaboration for Skill Learning.** In scenarios where robots teach skills, such as dance training and home robot guidance, comfort prediction is essential. It provides cognitive and kinesthetic feedback, making the skill acquisition process more responsive to user performance. **Real-Time Path Adjustment in Remote Control.** In teleoperation contexts, predicting human motion intentions and aligning intentions between humans and robots are crucial. Comfort prediction can enable real-time trajectory adjustments for robots, ensuring smooth and intuitive human-robot collaboration, especially in tasks requiring coordination.

5 LIMITATIONS AND FUTURE WORK

The study gauged comfort using GSR, but overlooked other physiological and psychological factors [4]. Future work will incorporate more biological and psychological data for a comprehensive comfort assessment. We aim to use probabilistic methods to capture variations from GSR signals, enhancing model robustness. This study only examined three interaction methods without considering scene influences or physiological data differences across methods. Future research will investigate more methods and contextual factors. This study didn't explore how robots can adapt to human interaction methods. We plan to use pre-trained language models like BERT, GPT-2, and GPT-3 to improve robot interactions.

6 CONCLUSION

We measured biometric data with a GSR wristband to detect discomfort in AR-enhanced human-robot interaction (AR-HRI). We used these data to train a model that can predict and alert discomfort, promoting more comfortable interactions. Our research aims to promote ergonomic and healthy designs for AR and interactive computer graphics. We also raise some open questions, such as how to design and evaluate comfortable interaction methods before deployment, and how to balance comfort and performance in immersive human-robot interaction. Our model can be used for ergonomic-aware optimization and innovation in AR-HRI.

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Table 1: Performance of the applied classifiers.

Method	Tr. prec.	Tr. F1	Te. prec.	Te. F1
SimpleRNN	0.6085	0.6081	0.5925	0.5916
LSTM	0.6718	0.5264	0.5879	0.4811
FCN	0.7750	0.7048	0.7126	0.6493
KNN	0.9050	0.7974	0.8420	0.7493